

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Design Webpage using HTML5 & CSS3

2. AIM

To design a static webpage using HTML5 semantic components and CSS3 styling features, including custom selectors, shadows, margins, borders, buttons, and layout alignments like Flexbox.

3. OBJECTIVE

The objectives of this experiment are:

- To understand semantic HTML5 structural components like headers, navigation bars, and section partitions.
 - To apply CSS3 internal styling rules to customize layout elements and typographical styles.
 - To implement visual styling enhancements using borders, corner rounding (border-radius), and drop shadows.
 - To utilize Flexbox alignment properties (display: flex, gap, justify-content, flex-wrap) to create a modular layout grid.
 - To construct responsive user interface elements including navigation bars and formatted card grids.
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4. THEORY

HTML5 introduces semantic tags (such as <header>, <nav>, <footer>, <section>, and <article>) which define document structure explicitly, enhancing readability for developers, search engines, and screen readers. In tandem, CSS3 (Cascading Style Sheets, Level 3) enables powerful styling capabilities, allowing the styling of layout regions, borders, shadows, alignment, and hover effects. Together, HTML5 and CSS3 support modern styling paradigms, rendering complex layouts natively without requiring third-party libraries.

Theory of HTML5 layout & CSS3 Styling:

- **CSS Flexbox Layout:** Flexbox (Flexible Box Layout Model) arranges page components dynamically. By using *display: flex*, alignment is managed across axes. Custom attributes include *justify-content* (e.g. *flex-end* to push links right or *center* to center room card grids), *gap* to establish gutters without margin hacks, and *flex-wrap: wrap* to push room boxes onto new lines when browser size shrinks, establishing native mobile responsiveness.
 - **CSS Box Model Properties:** Visual boxes are customized using *padding* to separate content from borders, *border* and *border-radius: 10px* for soft corner definitions, and *box-shadow: 2px 2px 10px rgba(0,0,0,0.1)* to apply soft transparent drop shadows that add page depth.
 - **CSS Typographical Selectors:** Tags (like *a*) are modified to remove underlines (*text-decoration: none*) and lists (*.navlist*) are styled to clear bullet symbols (*list-style-type: none*). Monospace and sans-serif fonts are customized via *font-family*.
 - **HTML5 Layout Structures:** The page uses `<header>` to contain title banners, semantic `<nav>` for navigation, and dividers (`<div class="room-box">`) to act as grid card elements containing images, captions, and buttons.
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5. ALGORITHM / PROCEDURE

1. Set up directories: *Src* (source), *Docs* (documentation), and *media/media* (image assets).
 2. Create *Static Webpage.html* inside *Src* and define basic HTML5 template structure.
 3. Configure metadata (viewport and character set) and define internal CSS styles under `<style>`.
 4. Design header container (`<header>`) with centered text and navigation links using standard lists.
 5. Apply CSS3 internal rules to style navigation list, aligning items with Flexbox and hiding anchor underlines.
 6. Construct a centered Flexbox container (*.room-container*) with responsive wraps (*flex-wrap: wrap*) and 20px gaps.
 7. Add room item cards (*.room-box*) containing local images, details, and styled booking buttons.
 8. Test webpage layout in browser, verifying Flexbox grid wrap adjustments, button font styles, and image displays.
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6. PROGRAM

File: *Static Webpage.html*

Static Webpage.html - Part 1 (Lines 1-50)

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Static webpage</title>
</head>
<style>
  .navlist{
    display: flex;
    gap: 25px;
    list-style-type: none;
    justify-content: flex-end;
  }
  a{
    text-decoration: none;
  }
  .room-container{
    display: flex;
    gap: 20px;
    justify-content: center;
    flex-wrap: wrap; /*Pushes the containers into next line on small screens*/
    padding: 50px;
  }
  .room-box{
    border: 2px solid #ddd;
    border-radius: 10px;
    padding: 15px;
    width: 500px;
    text-align: center;
    box-shadow: 2px 2px 10px rgba(0,0,0,0.1);
  }
  .room-box img{
    width: 100%;
  }
  .room-box button{
    color: blue;
    border: 2px solid #ddd;
    border-radius: 15px;
    width: 20%;
    height: 5%;
    font-family: 'Lucida Sans', 'Lucida Sans Regular', 'Lucida Grande', 'Lucida Sans Unicode', Geneva,
Verdana, sans-serif;
  }
</style>
<body>
  <header>
    <center>
```

Static Webpage.html - Part 2 (Lines 51-99)

```
<h1>THE STATIC WEBPAGE</h1>
</center>
<nav>
  <h3>THE KING BOOKINGS</h3>
  <ul class="navlist">
    <a href=""><li>HOME</li></a>
    <a href=""><li>HOTELS</li></a>
    <a href=""><li>BOOKINGS</li></a>
    <a href=""><li>ABOUT US</li></a>
  </ul>
</nav>
</header>
<h2 style="text-align: center;">THE ROOMS WE HAVE</h2><br>
<!-- The Flexbox grid for your rooms -->
<div class="room-container">

  <div class="room-box">
    <!-- Replace the src link with your actual image path -->
    
    <h3>Room 1</h3>
    <p>Big bed, AC, and premium vibes. ■2000/night.</p>
    <button>BOOK NOW</button>
  </div>

  <div class="room-box">
    
    <h3>Room 2</h3>
    <p>Ocean view with a crazy balcony. ■3500/night.</p>
    <button>BOOK NOW</button>
  </div>

  <div class="room-box">
    
    <h3>Room 3</h3>
    <p>Standard room. Doesn't break the bank. ■1200/night.</p>
    <button>BOOK NOW</button>
  </div>

  <div class="room-box">
    
    <h3>Room 4</h3>
    <p>VIP Suite. Strictly for the bosses. ■5000/night.</p>
    <button>BOOK NOW</button>
  </div>

</div>
</body>
</html>
```

7. CODE EXPLANATION

Code / Style Selector	Explanation
<code>.navlist { display: flex; gap: 25px; list-style-type: none; justify-content: flex-end; }</code>	Applies Flexbox layout to align navigation links horizontally to the right with 25px gaps, removing standard bullet markers.
<code>a { text-decoration: none; }</code>	Removes default blue underlines on anchor hyperlinks.
<code>.room-container { display: flex; gap: 20px; justify-content: center; flex-wrap: wrap; padding: 50px; }</code>	Enables Flexbox grid layout centering room cards with a 20px gap, wrapping cards automatically to the next line on smaller screens.
<code>.room-box { border: 2px solid #ddd; border-radius: 10px; padding: 15px; width: 500px; text-align: center; box-shadow: ... }</code>	Defines room card structure: styles border, adds 10px corner rounding, 15px padding, limits card width to 500px, and sets a soft shadow.
<code>.room-box button { color: blue; border: 2px solid #ddd; border-radius: 15px; width: 20%; font-family: ... }</code>	Styles standard booking buttons with custom borders, rounded corners, specific width/height, and Lucida Sans font.
<code><header> & <nav></code>	Semantic tags identifying heading and navigation content sections of the HTML page.
<code><div class="room-box"></code>	Represents a modular container block containing layout info for one room.
<code><button>BOOK NOW</button></code>	Renders the customized interactive action button for reserving room slots.

8. TEST CASES AND EXECUTION DATA

The following manual test cases were executed in Google Chrome to verify HTML5 layout and CSS3 styles:

TC ID	Test Description	Input / Action	Expected Result	Status
TC-01	Navigation Bar Alignment	Load page, inspect header	Title and navigation links align correctly; links align right	Pass
TC-02	Grid Item Layout	View room cards in browser	Four cards render in flex-grid layout with borders and shadows	Pass
TC-03	Image Render Paths	Open page and check room images	All local images load correctly inside card wrappers	Pass
TC-04	Flexbox Responsiveness	Narrow browser screen width	Room cards wrap onto new lines dynamically (flex-wrap: wrap)	Pass
TC-05	Button Visual Styles	Check booking buttons	Buttons render with rounded corners, blue text, and Lucida Sans font	Pass

9. OUTPUT

Output: Rendered Webpage in Chrome Browser

THE STATIC WEBPAGE

THE KING BOOKINGS

[HOME](#) [HOTELS](#) [BOOKINGS](#) [ABOUT US](#)

THE ROOMS WE HAVE



Room 1

Big bed, AC, and premium vibes. ₹2000/night.

[BOOK NOW](#)



Room 2

Ocean view with a crazy balcony. ₹3500/night.

[BOOK NOW](#)



Room 3

Standard room. Doesn't break the bank. ₹1200/night.

[BOOK NOW](#)



Room 4

VIP Suite. Strictly for the bosses. ₹5000/night.

[BOOK NOW](#)

10. RESULT

Thus, a static webpage incorporating HTML5 structural tags and CSS3 styling features (Flexbox alignments, box shadows, borders, rounded corners, custom text properties) was successfully designed, implemented, and verified using test cases.

Submission Package Structure:

Experiment2-Week2/

■ ■ ■ ■ Docs/

■ ■ ■ ■ Experiment 2.pdf

■ ■ ■ ■ Screenshots/

■ ■ ■ ■ screenshot-file-D-CSM259-...-2026-08-09-12_56_12.png

■ ■ ■ ■ Src/

■ ■ ■ ■ Static Webpage.html

■ ■ ■ ■ media/media/

■ ■ ■ ■ img1.jpg

■ ■ ■ ■ img2.jpg

■ ■ ■ ■ img3.jpg

■ ■ ■ ■ img4.jpg